

# VELMÈRE BASIC AUDIT REPORT

Lido Staked ETH Protocol (0xae7ab96520de3a18e5e111b5eaab095312d7fe84)

## VELMERE SECURITY AUDIT REPORT: LIDO STAKED ETH PROTOCOL

Network: Ethereum Mainnet (Chain ID: 1)

Report ID: undefined | Tier: BASIC | Application Surface: CANONICAL

Created At: 2026-09-09T19:29:41.983Z

VERDICT SUMMARY:

LOW RISK (24/100)

Release Decision: PASS | Verification Status: AUTOMATED\_ONLY

Confidence Score: 97/100

Evidence Coverage: 96%

Merkle Root: sha256:ac9c28cdfbd7d1a2bbfe9bbe5699d1360f60c9d6e85e1c717658dd1f59c70f10

stETH is the benchmark liquid staking derivative governed by Lido DAO with oracle consensus accounting rebase reports.

### AUDIT OVERVIEW & CONTRACT CONTEXT

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: Lido Staked ETH Protocol  
Contract Address: 0xae7ab96520de3a18e5e111b5eaab095312d7fe84  
Network / Chain ID: Ethereum Mainnet (ID: 1)  
Token / Symbol: stETH  
Proxy Pattern: AppProxyUpgradeability (Aragon DAO)  
Compiler Version: solc 0.8.9

### SOURCE CODE VERIFICATION & BYTECODE DECOMPIlation

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

### BASELINE SECURITY FINDINGS

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

#### Findings & Vulnerability Analysis:

LOW

VLM-LIDO-01: Rebasing Balance Dynamics

Category: Rebase Invariant | State: recommended

Evidence: function balanceOf(address \_account) public view returns (uint256)

Recommendation: Wrap into wstETH when collateralizing in automated lending vaults.

PERMISSION & ROLE GOVERNANCE MAP (PRO)PRO

[LOCKED] [LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

#### Summary of Analytical Depth in this section:

- Admin role: privilege and access control hierarchy in bytecode
- Verification of timelock delay parameters and governance controls
- Detection of privileged mint, pause, blacklist, and drainage capabilities
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER\_ENTITLEMENT\_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

LIQUIDITY HOLDER CONCENTRATION & LOCK EVIDENCE (PRO)PRO

[LOCKED] [LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

#### Summary of Analytical Depth in this section:

- Token distribution curve and top holder concentration analysis
- DEX liquidity pool pairing and lock verification across registered vaults
- Automated honeypot simulation, transfer fee checks, and sell restriction audit
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
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|                                                  |                                             |     |
|--------------------------------------------------|---------------------------------------------|-----|
| ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO) |                                             | PRO |
| [LOCKED]                                         | [LOCKED SECTION - REQUIRES PRO ENTITLEMENT] |     |

Summary of Analytical Depth in this section:

- Cross-function and cross-contract reentrancy guard modeling
- Price oracle staleness, manipulation resilience, and market feed verification
- Economic shock modeling and flash-loan manipulation resistance simulation
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER\_ENTITLEMENT\_BOUNDARY (client active tier does not cover this analytical layer).

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|                                                                   |                                                  |          |
|-------------------------------------------------------------------|--------------------------------------------------|----------|
| STORAGE LAYOUT COLLISION & MULTICOMPILER BYTECODE DIFF (ADVANCED) |                                                  | ADVANCED |
| [LOCKED]                                                          | [LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT] |          |

Summary of Analytical Depth in this section:

- Storage slot allocation collision detection across implementation upgrades
- Upgrade proxy layout gap verification and state preservation checks
- Bytecode reproduction against official compiler metadata and build pipelines
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
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|                                                              |                                                  |          |
|--------------------------------------------------------------|--------------------------------------------------|----------|
| FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED) |                                                  | ADVANCED |
| [LOCKED]                                                     | [LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT] |          |

Summary of Analytical Depth in this section:

- SMT proofs for solvency, conservation, and reentrancy impossibility invariants
- Bytecode regression analysis and logic change tracking
- Formal SMT Verification (Z3): PROPERTY P VERIFIED UNDER ASSUMPTIONS A FOR MODEL M
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER\_ENTITLEMENT\_BOUNDARY (client active tier does not cover this analytical layer).

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|                                                                |                                                  |          |
|----------------------------------------------------------------|--------------------------------------------------|----------|
| HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED) |                                                  | ADVANCED |
| [LOCKED]                                                       | [LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT] |          |

Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed attestation digest
- Manual review boundary explicit: no certified safe claims
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
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